Self-Learning: Topological Sorting Data Structures

> **Definition: Topological Sorting** of a **Directed Acyclic Graph (DAG)** is a linear ordering of vertices such that for every directed edge **(u, v)**, vertex **u** comes strictly **before** vertex **v** in the ordering.

---

1. Detailed Technical Explanation

```

Sampl

Valid Topological Orders:

- 5 ->

### Key Conditions:

1. Topological sorting is defined **ONLY for DAGs** (Directed Acyclic Graphs).

2. If a

---

## 2. Algorithms for Topological Sorting

### 1. Kahn's Algorithm (BFS In-Degree Approach)

1. Compute the **In-Degree** of every vertex in the graph.

2. Pus

### 2. DFS with Stack Approach

1. Initialize a `visited[]` boolean array.

2. For

---

## 3. Real-World Applications
```

- **Build Systems (Make, Gradle, Webpack):** Resolving source code compilation dependencies.

- **T

---

4. Quick Recall Flow

```
DAG C
```